

IMS Hands-On Lab

z/OS Connect and IMS OpenAPI 2

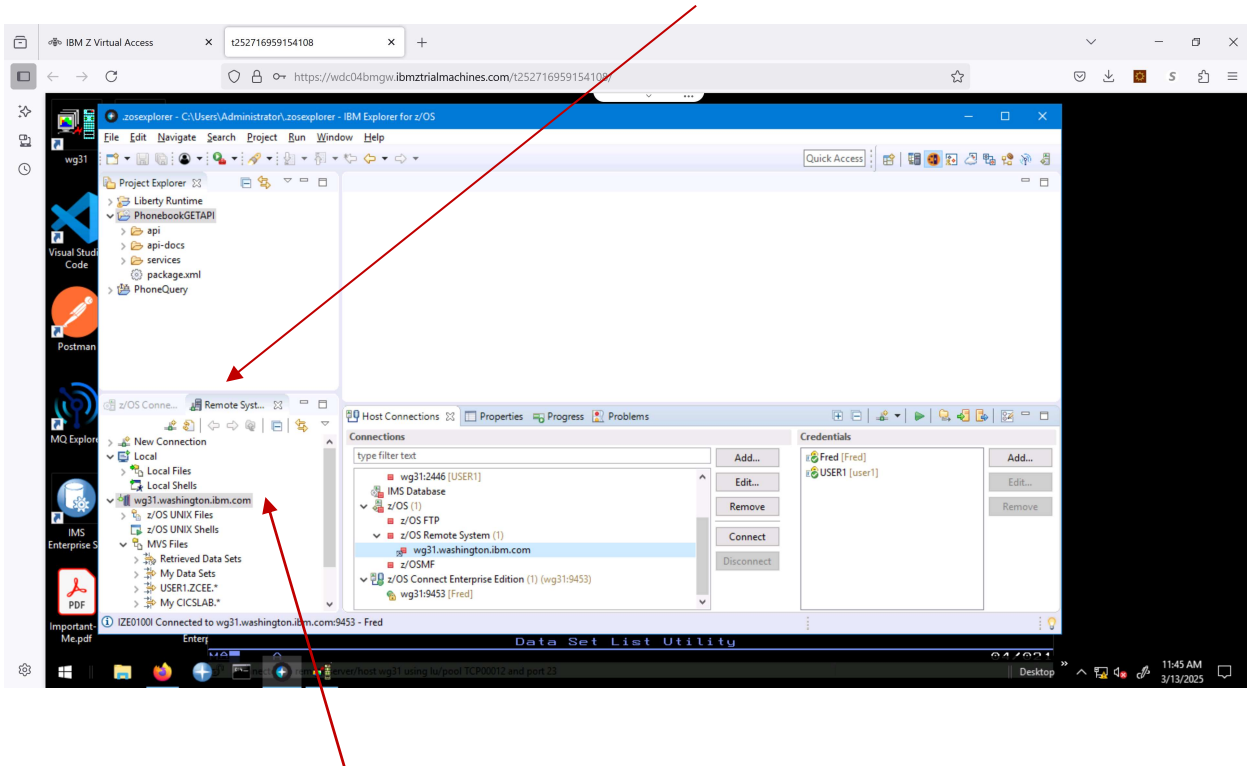
Troubleshooting

When problems occur, or if you need a closer look at the interactions that are occurring, there are several places which provide information,

- | | |
|---|--------------|
| 1. z/OS Connect environment traces/messages log | p. 2 |
| 2. Chec the z/OS Connect server in z/OS for errors | p. 7 |
| 3. Check the IMS control region for errors | p. 8 |
| 4. Check IMS Connect for errors | p. 9 |
| 5. Turn on the IMS Connect recorder trace | p. 10 |

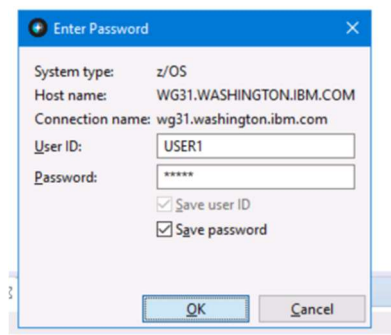
1. For the z/OS Connect environment, traces and message logs:

1-a. In z/OS Explorer, click on the Remote Server tab

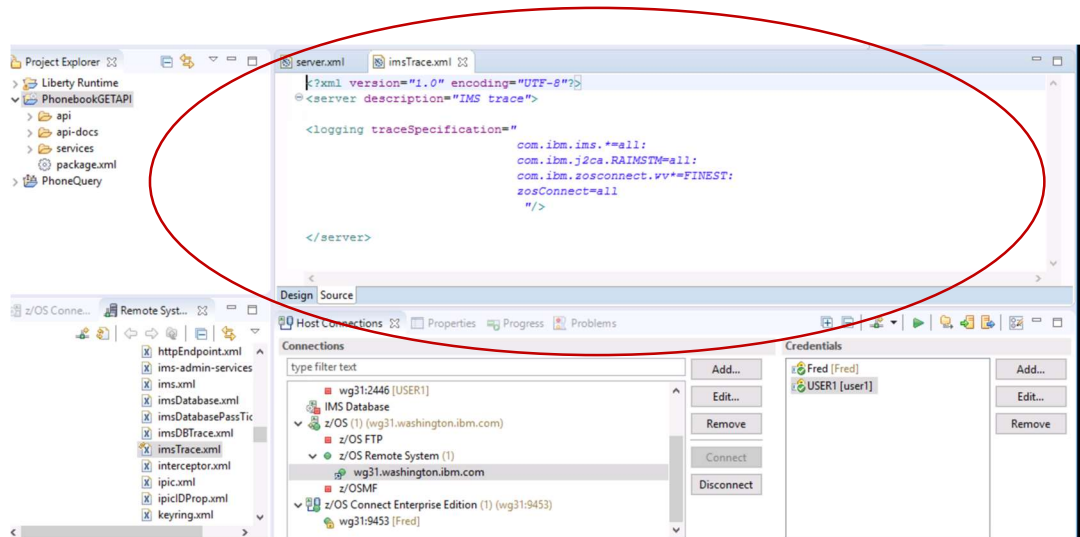


Expand wg31.washington.ibm.com

1-b. When challenged for userid/password, key in USER1 / USER1

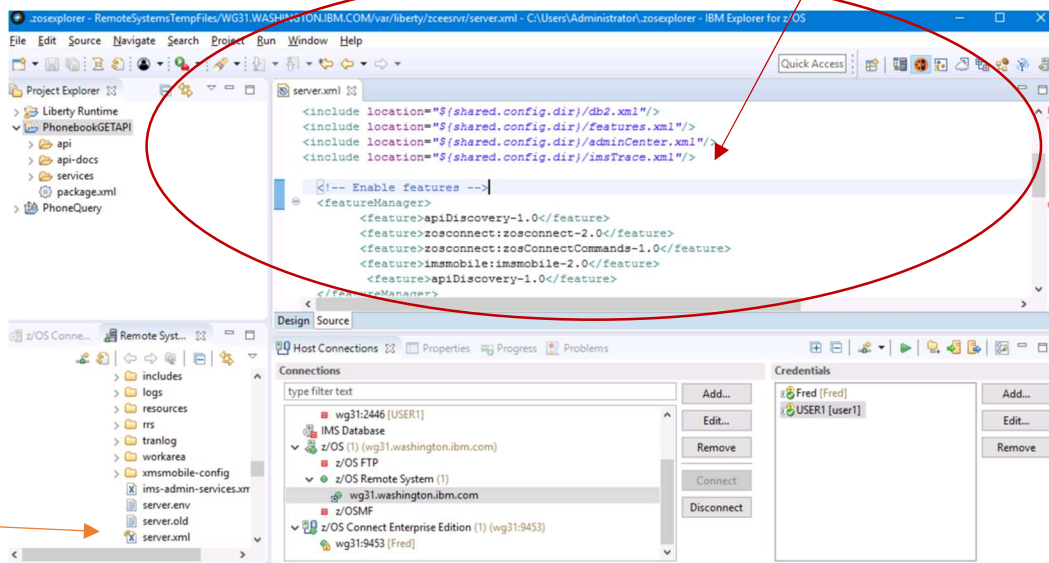


1-c. Follow the path: **z/OS Unix File > / > var /Liberty > zceesrvr** in the **includes > config** --- look at imsTraces.xml



1-d. This specification will need to be included in the server.xml file.

If this line does not exist in the server.xml file, add it in and save.



If the line was already there, then the trace is on, continue to the directions in 1-e,

Otherwise, make the change, save the server.xml file.

If you had to add the line, you will need to refresh the z/OS Connect server on the z/OS system.

- Log on to **TSO** using the USER1/USER1 id and password

- On the **ISPF** panel, key in **D.LOG**

```

Session A - [43 x 80]
File Edit Settings View Communication Actions Window Help
Menu Utilities Compilers Options Status Help

ISPF Primary Option Menu
Option ==>

0 Settings      Terminal and user parameters
1 View          Display source data or listings
2 Edit          Create or change source data
3 Utilities     Perform utility functions
4 Foreground    Interactive language processing
5 Batch         Submit job for language processing
6 Command       Enter TSO or Workstation commands
7 Dialog Test   Perform dialog testing
9 IBM Products  IBM program development products
10 SCLM         SW Configuration Library Manager
11 Workplace    ISPF Object/Action Workplace
D SDSF          Spool Search and Display Facility
M More          Additional IBM Products

Enter X to Terminate using log/list defaults

User ID . : USER1
Time . . : 19:13
Terminal . : 3278
Screen . . : 1
Language . : ENGLISH
Appl ID . : ISR
TSO logon . : ISPFPROC
TSO prefix : USER1
System ID . : SOW1
MVS acct. . : ACCT#
Release . . : ISPF 8.1
  
```

- This will open up SDSF SYSLOG panel where you can issue the command to refresh the ZCEESRVR
- Key in: **/f zceesrvr,refresh,config**

```

Session A - [43 x 80]
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help

SDSF SYSLOG 0.101 SOW1 SOW1 03/13/2025 2W DATA SET DISPLAYED
COMMAND INPUT ==> /f zceesrvr,refresh,config_ SCROLL ==> PAGE
RESPONSE=SOW1
+CWWKKB0005I: COMMAND RESPONSES COMPLETED SUCCESSFULLY FROM File
Refresh Command Handler.
02000000 SOW1 25072 11:42:47.42 STC06129 00000281 $HASP100 SMFCLEAR ON STO
00200000 SOW1 25072 11:42:47.43 STC06129 00000290 IEF695I START SMFCLEAR W
, GROUP SYS1
40000000 SOW1 25072 11:42:47.43 STC06129 00000090 $HASP373 SMFCLEAR STARTE
40000000 SOW1 25072 11:42:48.30 STC06129 00000281 $HASP395 SMFCLEAR ENDED
00000000 SOW1 25072 11:42:48.30 00000281 IEA989I SLIP TRAP ID=X33
00800000 SOW1 25072 11:45:56.71 00000090 IRR812I PROFILE * (G) IN
381 00000090 TO START BPXAS W
02000000 SOW1 25072 11:45:56.72 STC06130 00000281 $HASP100 BPXAS ON STO
40000000 SOW1 25072 11:45:56.74 STC06130 00000090 $HASP373 BPXAS STARTE
00000000 SOW1 25072 11:45:56.74 STC06130 00000290 BPXP024I BPXAS INITIATOR
0063
NC00000000 SOW1 25072 11:48:12.73 INSTREAM 00000290 LOGON
N 02000000 SOW1 25072 11:48:17.89 TSU06131 00000281 $HASP100 USER1 ON TSO
N 40000000 SOW1 25072 11:48:17.93 TSU06131 00000090 $HASP373 USER1 STARTE
  
```

- To View the server messages in the job... on the command input line, key in: **pre zceesrvr**

```

File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help

SDSF MENU 3.1 ADCDPL SOW1 LINE 1-38 (98)
COMMAND INPUT ==> pre zceesrvr_ SCROLL ==> P

NP NAME Description Group Status
DA Active users Jobs
I Input queue Jobs
O Output queue Output
H Held output queue Output
ST Status of jobs Jobs
JG Job groups JES
SYM System symbols System
LOG System log Log
SR System requests Log
MAS Members in the MAS JES
JC Job classes JES
SE Scheduling environments WLM
RES WLM resources WLM
ENC Enclaves WLM
PS Processes OMVS
  
```

- And then an **st** to see the jobs that match the prefix you entered.

```

Display Filter View Print Options Search Help
-----
SDSF STATUS DISPLAY ALL CLASSES
COMMAND INPUT ==> st_
NP JOBNAME JobID Owner Prty Queue C Pos Saff ASys Status
ZCEESRVR STC07432 ATSSERV 15 EXECUTION
LINE 1-1 (1)
SCROLL ==>
SOW1 SOW1

```

- Select the active server

```

SDSF STATUS DISPLAY ALL CLASSES
COMMAND INPUT ==>
NP  JOBNAME  JobID  Owner  Prty Queue  C  Pos  SA
s_  ZCEESRVR STC07432 ATSSERV 15 EXECUTION

```

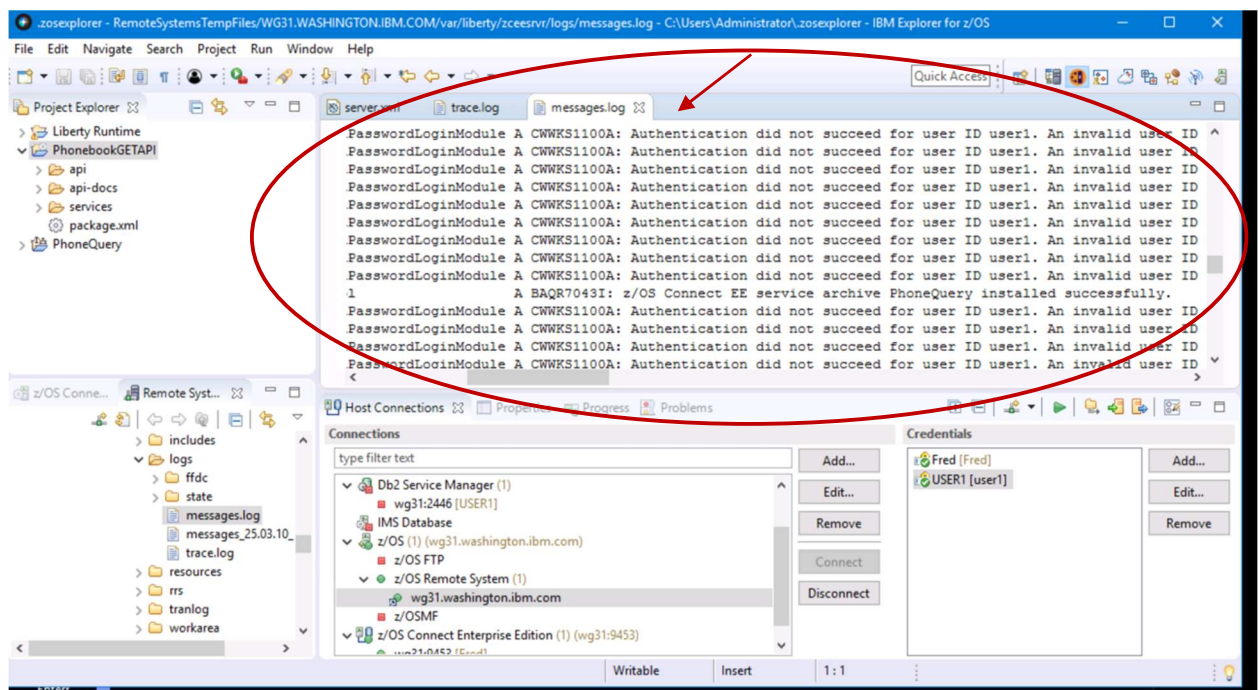
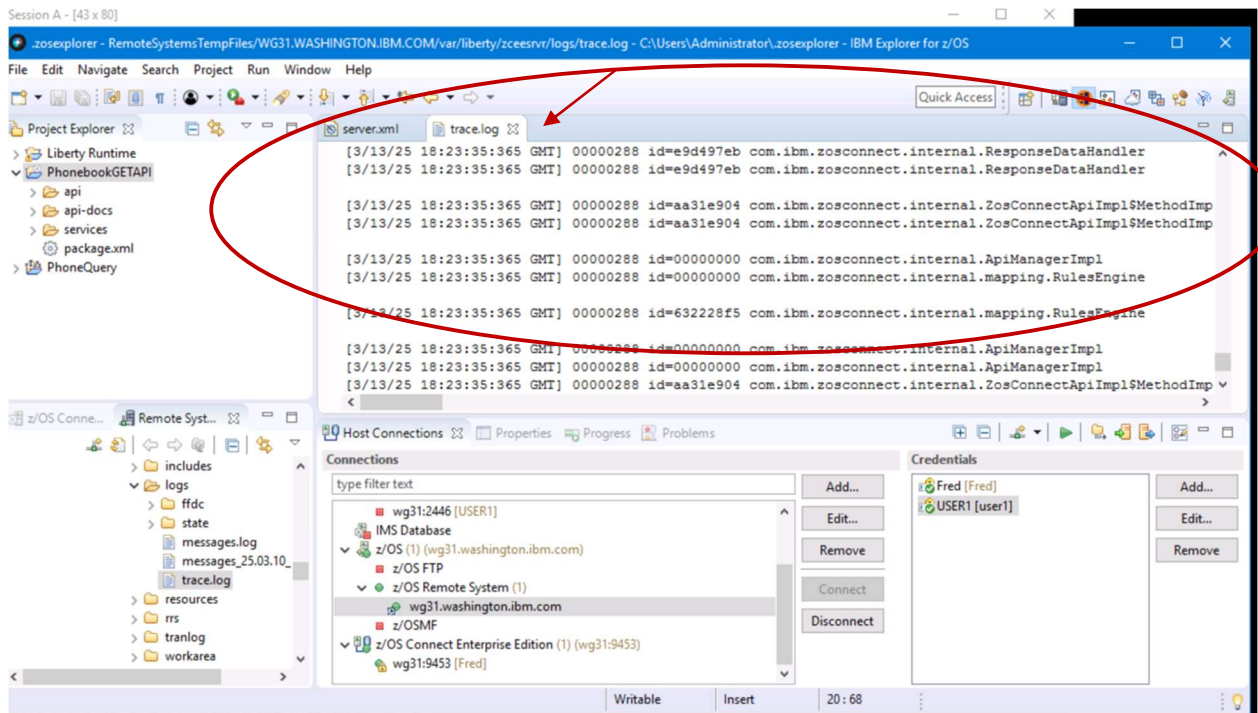
- At the bottom, you should see that the server configuration was updated

```

swordLoginModule A CWWKS1100A: Authentication did not succeed for user ID user1.
swordLoginModule A CWWKS1100A: Authentication did not succeed for user ID user1.
swordLoginModule A CWWKS1100A: Authentication did not succeed for user ID user1.
swordLoginModule A CWWKS1100A: Authentication did not succeed for user ID user1.
swordLoginModule A CWWKS1100A: Authentication did not succeed for user ID user1.
er A CWWKG0016I: Starting server configuration update.
er A CWWKG0028A: Processing included configuration resource: /var/
er A CWWKG0028A: Processing included configuration resource: /var/
er A CWWKG0028A: Processing included configuration resource: /var/
er A CWWKG0028A: Processing included configuration resource: /var/
er A CWWKG0028A: Processing included configuration resource: /var/
er A CWWKG0028A: Processing included configuration resource: /var/
er A CWWKG0028A: Processing included configuration resource: /var/
er A CWWKG0028A: Processing included configuration resource: /var/
er A CWWKG0028A: Processing included configuration resource: /var/
er A CWWKG0028A: Processing included configuration resource: /var/
er A CWWKG0028A: Processing included configuration resource: /var/
er A CWWKG0028A: Processing included configuration resource: /var/
er A CWWKG0028A: Processing included configuration resource: /var/
on I TRAS0018I: The trace state has been changed. The new trace st
er A CWWKG0017I: The server configuration was successfully updated
***** BOTTOM OF DATA *****

```


- 1- e. Once the server is available with trace specifications, then while the system is running both messages and trace information can be reviewed.



2. You can also check the messages on the z/OS Connect server running in z/OS.

- Log on to TSO and on the ISPF panel, select **D** for SDSF.
- Key in pre zceesrvr

```
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
-----
SDSF MENU 3.1      ADCDPL      SOW1      LINE 1-38 (98)
COMMAND INPUT ==> pre zceesrvr_      SCROLL ==> P
NP  NAME      Description      Group      Status
DA  Active users      Jobs
I  Input queue      Jobs
O  Output queue      Output
H  Held output queue      Output
ST  Status of jobs      Jobs
JG  Job groups      JES
SYM  System symbols      System
LOG  System log      Log
SR  System requests      Log
MAS  Members in the MAS      JES
JC  Job classes      JES
SE  Scheduling environments      WLM
RES  WLM resources      WLM
ENC  Enclaves      WLM
PS  Processes      OMVS
```

```
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
-----
SDSF STATUS DISPLAY ALL CLASSES      LINE 1-1 (1)
COMMAND INPUT ==> st_      SCROLL ==>
NP  JOBNAME JobID Owner Prty Queue C Pos SAff ASys Status
ZCEESRVR STC07432 ATSSERV 15 EXECUTION SOW1 SOW1
```

```
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
-----
SDSF STATUS DISPLAY ALL CLASSES      LINE
COMMAND INPUT ==>
NP  JOBNAME JobID Owner Prty Queue C Pos SAff ASys Status
S_  ZCEESRVR STC07432 ATSSERV 15 EXECUTION SOW1 SOW1
```

- Alternatively, on the command input line, you can key in: **DA** and put an **S** in front of the address space you want to review. In an active system, there may be many address spaces so the above directions get you to a specific address space.
- You can view all the different messages that might indicate an error condition

```

swordLoginModule A CWWKS1100A: Authentication did not succeed for user ID user1.
swordLoginModule A CWWKS1100A: Authentication did not succeed for user ID user1.
swordLoginModule A CWWKS1100A: Authentication did not succeed for user ID user1.
swordLoginModule A CWWKS1100A: Authentication did not succeed for user ID user1.
swordLoginModule A CWWKS1100A: Authentication did not succeed for user ID user1.
eRequestListener W BAQR0405W: The asynchronous request under URL https://wg31:94
A J2CA0056I: The Connection Manager received a fatal connection
Servlet W BAQR0429W: API testphonebookgetapi encountered an error while
eRequestListener W BAQR0405W: The asynchronous request under URL https://wg31:94
A J2CA0056I: The Connection Manager received a fatal connection
Servlet W BAQR0429W: API testphonebookgetapi encountered an error while
swordLoginModule A CWWKS1100A: Authentication did not succeed for user ID USER1.
swordLoginModule A CWWKS1100A: Authentication did not succeed for user ID USER1.
eRequestListener W BAQR0405W: The asynchronous request under URL https://wg31:94
A J2CA0056I: The Connection Manager received a fatal connection
Servlet W BAQR0429W: API testphonebookgetapi encountered an error while
***** BOTTOM OF DATA *****

```

3. To view the IMS Control region for error messages or to review the environment

- Once again, in TSO – on the ISPF panel, select **D** for SDSF.
- Key in: **pre IMS15CR1** on the command input line
- Key in: ST
- Select the IMS15CR1 server
 - Alternatively, on the command input line, you can key in: **DA** and put an **S** in front of IMS15CR1

```

SDSF OUTPUT DISPLAY IMS15CR1 STC07437 DSID 103 LINE 57 COLUMNS 01- 80
COMMAND INPUT ==> -DBRC LINE 001 PTERM 001. SCROLL ==> PAGE
IMS15DL1 DLS LINE 001 PTERM 001.
VTAM ACB OPEN -LOGONS ENABLED LINE 001 PTERM 001.
IMSLU=N/A,N/A APPC STATUS=DISABLED TIMEOUT= 0 MAXC= 5000 LIN
OTMA GROUP=OTMAGRP STATUS=ACTIVE LINE 001 PTERM 001.
APPC/OTMA SHARED QUEUE STATUS - LOCAL=INACTIVE GLOBAL=INACTIVE
APPC/OTMA SHARED QUEUES LOGGING=N LINE 001 PTERM 001.
APPC/OTMA RRS MAX TCBS - 40 ATTACHED TCBS - 2 QUEUED RRSWKS- 0 L
APPLID=IMS15 GRSNAME= STATUS=DISABLED LINE 001 P
TCP/IP.GENIMSID= STATUS=DISABLED LINE 001 PTERM 001.
LINE ACTIVE-IN - 1 ACTIV-OUT - 0 LINE 001 PTERM 001.
NODE ACTIVE-IN - 0 ACTIV-OUT - 0 LINE 001 PTERM 001.
*25100/105154* LINE 001 PTERM 001.
/DIS TRAN IVTNO LINE 001 PTERM 001.
TRAN CLS ENQCT QCT LCT PLCT CP NP LP SEGSZ SEGNO PARLM RC LI
IVTNO 1 0 0 65535 65535 1 1 1 0 0 NONE 0 LIN
PSBNAME. DFSIVP1 LINE 001 PTERM 001.
*25100/105442* LINE 001 PTERM 001.
DFS2500I DATABASE IVPDB1I SUCCESSFULLY ALLOCATED .
DFS2500I DATABASE IVPDB1 SUCCESSFULLY ALLOCATED .
DFS2864I EXTERNAL TRACE DATASET DFSTRA02 FULL - SWITCHING TO DFSTRA01 ..
DFS970I 14.04.41 UNEXPECTED STATUS ,NODE TCP00005,USER N/A ,***** ,RC=0
,FDB2=00,LOSTERM=14,SENSE=00000000,REASON=00.
DFS970I 14.04.41 CLSDST/TRMSESS FAILED,NODE TCP00005,USER N/A ,CLSDST,RC=1
,FDB2=13,LOSTERM=00,SENSE=00000000,REASON=00.
DFS970I 14.06.23 UNEXPECTED STATUS ,NODE TCP00004,USER N/A ,***** ,RC=0
,FDB2=00,LOSTERM=14,SENSE=00000000,REASON=00.
DFS970I 14.06.23 CLSDST/TRMSESS FAILED,NODE TCP00004,USER N/A ,CLSDST,RC=1
,FDB2=13,LOSTERM=00,SENSE=00000000,REASON=00.
/DIS TRAN IVTNO LINE 001 PTERM 001.
TRAN CLS ENQCT QCT LCT PLCT CP NP LP SEGSZ SEGNO PARLM RC LI
IVTNO 1 23 12 65535 65535 1 1 1 0 0 NONE 1 LIN
PSBNAME. DFSIVP1 LINE 001 PTERM 001.
*25100/141843* LINE 001 PTERM 001.
DFS970I 14.05.09 UNEXPECTED STATUS ,NODE TCP00003,USER USER1 ,***** ,RC=0
,FDB2=00,LOSTERM=14,SENSE=00000000,REASON=00.
DFS970I 14.19.05 CLSDST/TRMSESS FAILED,NODE TCP00003,USER N/A ,CLSDST,RC=1
,FDB2=13,LOSTERM=00,SENSE=00000000,REASON=00.
***** BOTTOM OF DATA *****

```


4. You can also view the IMS Connect for error messages or to review the environment

- If not already in SDSF, then on TSO – on the ISPF panel, select **D** for SDSF.
- Key in: **pre IMS15HWS1** on the command input line
- Key in: ST
- Select the IMS15HWS1 server
 - Alternatively, on the command input line, you can key in: **DA** and put an **S** in front of IMS15HWS

```

Session A - [43 x 80]
File Edit Settings View Communication Actions Window Help
-----
Display Filter View Print Options Search Help
-----
SDSF OUTPUT DISPLAY IMS15HWS STC05988 DSID 4 LINE 30 COLUMNS 02- 81
COMMAND INPUT ==> SCROLL ==> PAGE
-----
HWS00001I NUMSQC=3 WARNOSC=80% WARNINC=5%
HWS00001I RRS=Y STATUS=ACTIVE MULTIRTP= CASCADE=N
HWS00001I VERSION=V15 IP-ADDRESS=010.001.001.002
HWS00001I SUPER MEMBER NAME= CM0 ACK TOQ=
HWS00001I ADAPTER=N MAXCVRT=100 NUMCVRT=0
HWS00001I MAXLSSSZ=32767
HWS00001I ODBM AUTO CONNECTION=Y
HWS00001I ODBM TIMEOUT=6000
HWS00001I ODBM IMSPLEX MEMBER=IMS15HWS TARGET MEMBER=PLEX1
HWS00001I DATASTORE=IVP1 STATUS=ACTIVE
HWS00001I GROUP=OTMAGRP MEMBER=HWSMEM
HWS00001I TARGET MEMBER=OTMAMEM STATE=AVAIL
HWS00001I DEFAULT REROUTE NAME=HWS$DEF IMS VERSION=15.1
HWS00001I RACF APPL NAME=IMSAPPL MULTIRTP= CASCADE=
HWS00001I OTMA ACCE AGING VALUE=999999 SENDALTP= TRANORCH=
HWS00001I OTMA ACK TIMEOUT VALUE=120
HWS00001I OTMA MAX INPUT MESSAGE=5000
HWS00001I SUPER MEMBER NAME= CM0 ACK TOQ=
HWS00001I IMSPLEX=PLEX1 STATUS=ACTIVE
HWS00001I MEMBER=IMS15HWS TARGET=PLEX1
HWS00001I ODBM=IVP1500D STATUS=REGISTERED ODBMRRS=N
HWS00001I ALIAS=IVP1 STATUS=ACTIVE
HWS00001I NO ACTIVE MSC
HWS00001I NO ACTIVE ISC
HWS00001I PORT=4000 STATUS=ACTIVE KEEPAV=0 NUMSQC=2 EDIT= TIM
HWS00001I CLIENTID USERID TRANCODE DATASTORE STATUS SECOND CLNTPORT
HWS00001I GMPKWKTF USER2 IVTNO IVP1 RECV 1559 16701 1
HWS00001I TOTAL CLIENTS=1 RECV=1 READ=0 CONN=0 XMIT=0 OTHER=0
HWS00001I PORT=5555D STATUS=ACTIVE KEEPAV=0 NUMSQC=1 EDIT= TIM

```

5. IMS connect can be configured with a file that records activity by providing an internal trace . This is called the **IMS Connect recorder file**.

- To enable this capability, the IMS Connect startup JCL must contain a specification for the sequential file.
- The DD is HWSRCORD

```

Menu Utilities Compilers Help
BROWSE DFSF10.PROCLIB(IMS15HWS) - 01.00 Line 0000000000 Col 001 080
Command ==> Scroll ==> PAGE
***** Top of Data *****
I IVP IMS 15.1
SKELETON: DFSIXSTG
FUNCTION: START IMS CONNECT REGION.
*****@SCPYRT**
LICENSED MATERIALS - PROPERTY OF IBM
5635-A06
COPYRIGHT IBM CORP. 1989,1998 ALL RIGHTS RESERVED
US GOVERNMENT USERS RESTRICTED RIGHTS - USE, DUPLICATION OR
DISCLOSURE RESTRICTED BY GSA ADP SCHEDULE CONTRACT WITH
IBM CORP.
*****@ECPYRT**
IMS15HWS PROC RGN=512M,TME=1440,SOUT=*,
BPECFG=BPECFGIV,HWSCFG=HWSCFODB
HWSREGN EXEC PGM=HWSHWS00,REGION=&RGN,TIME=&TME,
PARM='BPECFG=&BPECFG,HWSCFG=&HWSCFG'
STEPLIB DD DSN=DFSF10.SDFSRESL,DISP=SHR
DD DSN=CEE.SCEERUN,UNIT=SYSALLDA,DISP=SHR
PROCLIB DD DSN=DFSF10.PROCLIB,DISP=SHR
SYSPRINT DD SYSOUT=&SOUT
SYSUDUMP DD SYSOUT=&SOUT
HWSRCORD DD DSN=DFSF10.HWSRCORD,DISP=SHR
PEND
HWS01 EXEC IMS15HWS

```

General Data		Current Allocation	
Management class	: STANDARD	Allocated cylinders	: 1
Storage class	: IMSDB	Allocated extents	: 1
Volume serial	: DB0001		
Device type	: 3390		
Data class	: **None**		
Organization	: NONE	Current Utilization	
Record format	: VB	Used cylinders	: 0
Record length	: 22524	Used extents	: 0
Block size	: 22528		
1st extent cylinders	: 1		
Secondary cylinders	: 0		
Data set name type	:	Dates	
Data set encryption	: NO	Creation date	: 2013/12/10
		Referenced date	: 2023/02/09
		Expiration date	: ***None***

5-a. to turn on the recorder trace for IMS Connect, you need to respond to the outstanding prompt for the IMS Connect address space. If you are not already in SDSF, then from the main ISPF menu, key in **D.LOG**

```

Session A - [43 x 80]
File Edit Settings View Communication Actions Window Help
Menu Utilities Compilers Options Status Help
ISPF Primary Option Menu
Option ==>
0 Settings      Terminal and user parameters      User ID . : USER1
1 View          Display source data or listings    Time .   : 19:13
2 Edit          Create or change source data       Terminal: 3278
3 Utilities     Perform utility functions       Screen . : 1
4 Foreground    Interactive language processing Language. : ENGLISH
5 Batch         Submit job for language processing Appl ID . : ISR
6 Command       Enter TSO or Workstation commands  TSO logon: ISPFPROC
7 Dialog Test   Perform dialog testing            TSO prefix: USER1
9 IBM Products  IBM program development products  System ID : SOW1
10 SCLM         SM Configuration Library Manager MVS acct. : ACCT#
11 Workplace    ISPF Object/Action Workplace       Release . : ISPF 8.1
D SDSF         Spool Search and Display Facility
M More          Additional IBM Products

Enter X to Terminate using log/list defaults
  
```

5-b. Once in the SDSF syslog, respond to the outstanding WTOR prompt for IMS Connect,

/xxrecorder open (where xx is the outstanding prompt)

```

Session A - [43 x 80]
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
SDSF SYSLOG      0.101 SOW1 SOW1 04/10/2025 2W      7,124 COLUMNS 02- 81
COMMAND INPUT ==> /xxRECORDER OPEN_                SCROLL ==> PAGE
S 42000000 SOW1 25100 14:18:43.68 STC07437 00000090 PARLM RC IVP1
S DFS000I IVTNO 1
N 42000000 SOW1 25100 14:18:43.68 STC07437 00000090 NONE 1 IVP1
N 42000000 SOW1 25100 14:18:43.68 STC07437 00000090 DFS000I PSBNAME:
W 42000000 SOW1 25100 14:18:43.68 STC07437 00000090 DFS000I *25100/14184
NC00000000 SOW1 25100 14:19:05.00 USER1 00000290 *08 DFS996I *IMS READY*
NRFFFFFFF SOW1 25100 14:19:05.00 JOB07470 00000090 R 07 SUPPRESSED
N FFFFFFFF SOW1 25100 14:19:05.02 JOB07470 00000090 IEE600I REPLY TO 07 IS;S
N FFFFFFFF SOW1 25100 14:19:05.02 JOB07470 00000090 +STATUS II, DLI CALL = I
N FFFFFFFF SOW1 25100 14:19:05.02 JOB07470 00000090 +STATUS GE, DLI CALL = G
N FFFFFFFF SOW1 25100 14:19:05.02 JOB07470 00000090 +STATUS GE, DLI CALL = G
S 42000000 SOW1 14:18.43 STC07437 *08 DFS996I *IMS READY* IVP1
B0000000 SOW1 19.45.41 STC07461 *06 HWSC0000I *IMS CONNECT READY* IMS15HWS
***** BOTTOM OF DATA *****
  
```

You should see a response that the recorder was opened.

5-c. At this point, go back and execute your API. If API messages are successfully sent to IMS Connect then trace records will be on the recorder log.

5-d. Back in TSO, on the SDDSF Syslog screen, turn off the recorder trace. Respond to the new outstanding prompt

/xxrecorder close

```

Session A - [43 x 80]
File Edit Settings View Communication Actions Window Help
Display Filter View Print Options Search Help
SDSF SYSLOG 0.101 SOW1 SOW1 04/10/2025 2W 7,124 COLUMNS 02- 81
COMMAND INPUT ==> /xxrecorder close SCROLL ==> PAGE
S N 4200000 SOW1 25100 14:18:43.68 STC07437 00000090 PARLM RC IVP1 1
S N 4200000 SOW1 25100 14:18:43.68 STC07437 00000090 DFS000I 1 IVP1
N 4200000 SOW1 25100 14:18:43.68 STC07437 00000090 DFS000I PSBNAME:
N 4200000 SOW1 25100 14:18:43.68 STC07437 00000090 DFS000I *25100/14184
W 4200000 SOW1 25100 14:18:43.68 STC07437 00000090 *08 DFS996I *IMS READY*
NC0000000 SOW1 25100 14:19:05.00 USER1 00000290 R 07 SUPPRESSED
NRFFFFFFF SOW1 25100 14:19:05.00 JOB07470 00000090 IEE600I REPLY TO 07 IS;S
N FFFFFFFF SOW1 25100 14:19:05.02 JOB07470 00000090 +STATUS II, DLI CALL = I
N FFFFFFFF SOW1 25100 14:19:05.02 JOB07470 00000090 +STATUS GE, DLI CALL = G
N FFFFFFFF SOW1 25100 14:19:05.02 JOB07470 00000090 +STATUS GE, DLI CALL = G
N FFFFFFFF SOW1 25100 14:19:05.02 STC07461 00000090 HWSP1485E PASSING TO TCP
S M=SDRC, ID=GMPT8T1F, IPV
NC0000000 SOW1 25100 14:25:51.12 USER1 00000290 R 06,RECORDER OPEN
NR8000000 SOW1 25100 14:25:51.12 STC07461 00000090 IEE600I REPLY TO 06 IS;R
W 8000000 SOW1 25100 14:25:51.12 STC07461 00000090 *09 HWSC0000I *IMS CONNEC
N FFFFFFFF SOW1 25100 14:25:51.14 STC07461 00000090 HWSR0880I RECORDER OPEN
8000000 SOW1 14.25.51 STC07461 *09 HWSC0000I *IMS CONNECT READY* IMS15HWS
4200000 SOW1 14.18.43 STC07437 *08 DFS996I *IMS READY* IVP1

```

5-e. You can now view the recorder dataset. As you saw in the earlier screenshot, the recorder dataset on this system is DFSF10.HWSRCDR.

```

File Edit Settings View Communication Actions Window Help
Menu RefList RefMode Utilities Help
Data Set List Utility 2 Members processed
Option ==>
blank Display data set list P Print data set list
V Display VTOC information PV Print VTOC information
Enter one or both of the parameters below:
Dsname Level . . . DFSF10.hwsrldr
Volume serial . . .
Data set list options
Initial View Enter "/" to select option
1. Volume / Confirm Data Set Delete
2. Space / Confirm Member Delete
3. Attrib / Include Additional Qualifiers
4. Total / Display Catalog Name
- Display Total Tracks
- Prefix Dsname Level
When the data set list is displayed, enter either:
"/" on the data set list command field for the command prompt pop-up,
an ISPF line command, the name of a TSO command, CLIST, or REXX exec, or
"=" to execute the previous command.

```

Browse the dataset


```

BROWSE      DFSF10.HWSRCDR                      Line 0000000000 Col 001 080
Command ==> hex ON                               Scroll ==> PAGE
***** Top of Data *****
....ICONRC.ÿ.....ÿ.....GMPC3S6J\½..!psî\½..$.îî.....
....ICONSN.ÿ.....ÿ.....GMPC3S6J\½..$°dî\½..$_°î.....
....ICONRC.ÿ.....ÿ.....GMPC3S6J\½..°j.î\½..°î.....
....ICONSN.ÿ.....ÿ.....GMPC3S6J\½..0.Fî\½..0.}î.....
***** Bottom of Data *****

```

The ICONRC record shows what IMS connect received

The ICONSN record shows the response from IMS

The following links describe how to read the records in detail

<https://www.ibm.com/docs/en/ims/15.5.0?topic=traces-recorder-log-record-mappings>

<https://www.ibm.com/docs/en/ims/15.5.0?topic=tailoring-jcl-print-ims-connect-recorder-output>

At this point, however, you can get an indication of the messages sent back and forth by scrolling to the right. As you can see, the first line shows the input to transaction IVTNO. The second line is the response, etc.

```

File Edit Settings View Communication Actions Window Help
Menu Utilities Compilers Help
BROWSE      DFSF10.HWSRCDR                      Line 0000000000 Col 641 720
Command ==>                               Scroll ==> PAGE
***** Top of Data *****
W1 \mB...)ÿp.....IVTNO      DISPLAY LAST1
PLAYED \mB...ÿ.....IVTNO      DISPLAY LAST1      FIRST1 8-111-1111D01/R010013...
W1 \mB...ÿ.....IVTNO      DISPLAY MOUSE
SON WAS NOT FOUND
***** Bottom of Data *****

```